

Mid-Term Syllabus

Course Code-MAT3003

Course Title- Probability, Statistics & Reliability

Module-1

Probability Theory:

Introduction to probability concepts, Random experiments, Events, Conditional probability, Independent events, Theorem of Total Probability, Baye's theorem, Sample space.

Module-2

Random variables (RV):

Introduction to Random variables (One-~~two~~ dimension) - Discrete and Continuous RV - Probability function and Distribution function of RV, Expectation, Variance, and its properties, Moment Generating function upto second moments.

Special Distributions:

Bernoulli distribution, Binomial distribution, Poisson distributions.

Module-3

Bivariate Random Variables:

Joint density function, joint probability distribution, Marginal Probability Distribution.

Only discrete case